

An applied scientific discipline concerned with how humans interact with the tools and equipment they use while performing tasks and other activities. Also called “human factors engineering”. Ergonomics seeks to develop a "fit" between people and the jobs they do.

Objectives in Ergonomics:

Greater ease of interaction between user and machine• Avoid errors and mistakes• Greater comfort and satisfaction in use of the equipment• Reduce stress and fatigue• Greater efficiency and productivity• Safer operation• Avoid accidents and injuries•

Introduction

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Physiology

Capacity of human body to use energy and apply forces depends on:

Capacity of cardiovascular and respiratory systems to deliver required fuel and oxygen to muscles and carry away waste products

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Muscle strength and endurance• Ability to maintain proper heat balance within the body•

Metabolism

Sum of the biochemical reactions that occur in the cells of living organisms Functions:

Provide energy for vital processes and activities, including muscle contraction1. Assimilate new organic material into the body2.

Basal metabolism

energy used only to sustain the vital circulatory and respiratory functions

Basal Metabolic Rate (BMR):

for a 20yr old male – 1.0 kcal/hr/kg.body weight

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for a 20yr old female – 0.9 kcal/hr/kg.body weight

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decreases slightly with age – 2% per decade

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Activity metabolism

energy associated with physical activity

Metabolic Rate (AMR): Varies significantly with activity

Digestive metabolism

energy used for digestion Digestive Metabolic Rate (DMR) is about 10% of (BMR+AMR)

Total daily metabolic rate

The following table shows typical energy expenditure for a person weighing 72Kg. If a person's weight is W, multiply ER by

Activity Energy Expenditure (ER)

Sleeping BMRm

Resting Seated 1.5 kcal/min

Standing (not walking) 2.2 kcal/min

Physical ergonomics

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Walking at 4.5 km/hr 4.0 kcal/min

Jogging at 7.2 km/hr 7.5 kcal/min

Mowing lawn (push mower) 8.3 kcal/min

Chopping wood 8.0 kcal/min

Office work seated 1.6 kcal/min

Light assembly work seated 2.2 kcal/min

Bricklaying 4.0 kcal/min

Shovelling coal 8.5 kcal/min

Energy expenditure

Energy expenditure, heart rate, and oxygen consumption for several categories of work activity.

Mean Energy Expenditure Rate:

Physiological measure Male worker

Female worker

Energy expenditure rate of the physical activity (maximum time-weighted average during shift)

5.0 kcal/min

4.0 kcal/min

Energy expenditure of the physical activity for the entire 8 hr shift

2400 kcal 1920 kcal

Heart rate (maximum time-weighted average during shift) 120 beats/min 110 beats/min

Oxygen debt

Both the recommended mean energy expenditure and oxygen debt show the importance of taking breaks

A 72 kg. male worker performs a repetitive task with a 12- min work cycle. During each cycle his energy expenditure rate is 7.8 kcal/min for 20% of the time and 5.6 kcal/min for the remaining 80%. On average, what rest break should be allowed at the end of each work cycle.

During task Average

Recommended max mean energy expenditure over 8 hour shift = 5.0kcal/min

Study of the capabilities and limitations of the human brain and sensory system while performing activities that have a significant information processing content

Sensing

When to use visual When to use auditory

Long message• Complex message• Referred to later• Noisy environment• Person receiving the message remains in one location

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Short message• Simple message• Requires action now• Very light or very dark• Person expected to move around• To attract attention to exceptions•

Guidelines:

Standardization - having similar devices operate in the same way•

Redundancy - presenting information using more than one sensory mode• Stimulus variation - use of variable stimuli rather than stimuli that are constant and continuous

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Graphical displays - information presented graphically usually more effective than same information presented as text

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Attention resources

Types of attention:

Selective attention

Monitor multiple sources of information to perceive irregularities or opportunities○

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Divided attention

Multiple stimuli, but multiple tasks to be performed together○

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Sustained attention

Watch for a signal of interest over a long time period, it is important to not miss the signal

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Focused attention

Cope with multiple input channels but focus on only one channel for a sustained time period

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Usually caused by boredom - state of being weary and restless due to lack of interest. Factors

Cognitive ergonomics

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leading to boredom on the job:

Short cycle times• Low requirements for body movements• Warm environment• Lack of contact with other workers• Low motivation• Low lighting levels in workplace•

Memory

Minimize demands on working memory:

minimize the number of alphanumeric items that must be kept in working memory and length of time they must be retained

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Exploit chunking:

Formulate meaningful sequences out of a string of alphanumeric characters so string can be retained as one chunk

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Use letters rather than numbers (letters are more likely to have meaning)• Limit chunk size to 3 or 4 characters•

Increase frequency and recency of using information stored in long-term memory:

Drills to recall emergency procedures• Regular and frequent training sessions• Standardized procedures in batch operations•

Use memory aids:

Written instructions for procedures that must be carried out in the correct sequence•

Mnemonics•

Execution

Decision complexity

As more choices become available, the complexity of the decision process increases, and the time to make the selection increases.

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Response expectancy

Humans can process information they are expecting much faster than information they are not expecting.

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They can select a response they are expecting to choose more quickly than one that is unexpected.

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Compatibility

Response compatibility means that the possible responses should be consistent with one's expectations.

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Trade-off between speed and accuracy

Negative correlation between speed and accuracy in response selection and execution•

Feedback

Allows a person to see and/or hear the effect of their actions. •

Design controls that obey normal conventions, and are intuitive

Controlled element Control Action

On Down, right, forward, clockwise

Off Up, left, backward

Right Right, clockwise

Raise Up, back

Lower Down, forward, up

Retract Backward, pull

Extend Down, forward, push

Increase Up, right, forward, clockwise

Decrease Down, left, backward, anticlockwise

Use coding to make controls easier to distinguish. Code by:

Location. e.g., car foot pedals• Colour. e.g., emergency stop button• Size and shape. e.g., aircraft cockpit controls• Labelling (text or icon). e.g., CNC machine tool•

The Visual Environment and Lighting:

About 80% of the information input to the human brain comes from visual stimuli.•

The Auditory Environment and Noise

The second most important means by which humans receive information. Visual and auditory stimuli account for 95% or more of a person's information input

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The auditory environment also includes: Noise - unwanted sound•

Climate Control in the Work Environment

Has significant effect on the physiological and cognitive processes involved in work•

Visual

Light - electromagnetic radiant energy that lies within the visible spectrum. Wavelengths between ~ 400 nm (blue-violet) and ~ 700 nm (red)

Visual acuity - capability to discriminate small objects or fine details

Visual angle

Where: h = height of object or detail d = distance from eye

Visual acuity:

Where: α_v is expressed in arc mins

Colour discrimination - capability to distinguish colours.

Approx 15-20% of population has some form of colour blindness.• Adaptation - ability to adapt to changes in light levels

Dark adaptation - adapting from a bright environment to a dark one• Light adaptation - adapting from a dark environment to a light one•

Luminous flux - rate at which light energy is emitted in all directions from a light source

Units: lumen (lm)

The physical work environment

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Luminous intensity - luminous flux emitted in a given direction Units: candela (cd)

Candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} Hz and that has a radiant intensity in that direction of $1/683$ watt per steradian. Hence a led isotropic point source of light radiates 4π lumens

Illuminance - luminous flux shining per unit area on a surface

Units: lux (lx) = 1 lumen per sq meter

Some products specify their beam angle θ

Illuminance decreases with distance from light source:

Where:

E = illuminance

I = luminous intensity

d = distance

When incident light is at an angle:

Illuminance Example

10

4 lux Total starlight, overcast sky

0.002 lux Moonless clear night sky with airglow

0.27 lux Full moon on a clear night

50 lux Family living room

100 lux Very dark overcast day

320–500 lux Office lighting

1,000 lux Overcast day; typical TV studio lighting

10,000–25,000 lux Full daylight (not direct sun)

32,000–130,000 lux Direct sunlight

Visibility depends on:

Visual angle • Level of illumination • Contrast • Colour •

Luminance (L) - amount of light reflected from a surface Units: cd/sq m

Depends on:

Illuminance - amount of light striking surface E • Reflectance - depends on colour and texture of surface R

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Object Typical Reflectance

Mirrored glass 0.85

White matt paint 0.82

Light green paint 0.65

Aluminium paint 0.65

Medium blue paint 0.35

Dark blue paint 0.08

Black paint 0.04

There are various ways of expressing contrast. Weber Contrast is expressed by the ratio:

There is an acceptable range, too low gives problems of distinguishing object, too high causes discomfort due to glare.

Auditory

Sound consists of pressure oscillations which are detected by the ear, which transforms mechanical energy of sound waves into electrical nerve signals that are transmitted to the brain for interpretation. Humans with normal hearing can perceive sound frequencies in the approximate range 20 Hz to 20,000 Hz when young. The aging process takes its toll; perception of high frequencies decreases with age. Hearing is also very prone to damage by extended exposure to high levels of sound. This has significant H&S implications.

Sound intensity is measured from the listener's perspective. It is not a power measurement of the sound source. Could be measured as pressure:

e.g., N/m² or Pa However, the range of human detectable sound pressures is very large (0.00002 N/m² to 20 N/m²). Therefore intensity is measured relative to a reference pressure and converted to

logarithmic scale called sound pressure level (SPL) with units of decibel (dB):

Where: p_s = sound pressure from source, N/m²

p_r = reference sound pressure, N/m² (the usual reference pressure is 0.00002 N/m²) Intensity of a sound wave varies inversely as the square of the distance from the source.

Threshold of hearing 0 dB

Soft whispering at 1 m (3 ft) 20 dB

Library environment 40 dB

Room air conditioner at 3 m (10 ft) 60 dB

Talking at 15 cm (6 in) 80 dB

Powered lawnmower at 1 m (3 ft) 100 dB

Punch Press at 1m 105 dB

Jet engine at 60 m (200 ft) 120 dB

Jet engine at 30 m (100 ft) 140 dB

Adding SPLs:

The Noise Regulations (2005) specify actions at certain levels of noise, when averaged over an 8 hour working day, and also specify exposure to maximum noise (peak sound pressure). Considering average noise:

Where: T_i is the time at SPL_i and T_0 is 8 hours

For daily exposure averaging above 80 dBA, employers must provide hearing protection if requested. For daily exposure averaging above 85 dBA, employers must provide hearing protection, and make sure it is used. Workers must not be exposed above 87 dBA, taking account of any reduction in exposure provided by hearing protection.

What the law says... (The Control of Noise at Work Regulations 2005) The lower exposure action values are

a daily or weekly personal noise exposure of 80 dBA; and a. a peak sound pressure of 135 dBA.b.

1.

The upper exposure action values are a daily or weekly personal noise exposure of 85 dBA; and a. a peak sound pressure of 137 dBA.b.

2.

The exposure limit values are a daily or weekly personal noise exposure of 87 dBA; and a. a peak sound pressure of 140 dBA.b.

3.

Where the exposure of an employee to noise varies markedly from day to day, an employer may use weekly personal noise exposure in place of daily personal noise exposure for the purpose of compliance with these Regulations.

4.

In applying the exposure limit values in (3), but not in applying the lower and upper exposure action values in (1) and (2), account shall be taken of the protection given to the employee by any personal hearing protectors provided by the employer

5.

A worker is exposed to two noise sources, one at 80 dBA and the other at 84 dBA.

Determine: a) the total sound pressure level of the two sources b) the permissible duration of exposure for this sound pressure level, to keep below the daily personal noise exposure level of 85dBA .

Startle response - due to sudden loud noise

Causes spontaneous muscle contractions, blinking eyes, headjerk movement • Hearing loss (three categories):

Temporary threshold shift - hearing impairment of short duration • Noise-induced permanent threshold shift - results from long term exposure to high noise levels

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Acoustic trauma - single exposure to high intensity noise can cause temporary or permanent hearing loss

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Administrative controls

Managing the exposure durations for employees working in noisy environments by setting time limits on exposure to noise level •

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Engineering controls

Noise abatement at three locations:

Source - design quieter machinery, insulation, vibration damping, lubrication, natural cooling instead of fans

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Receiver - use of ear plugs, helmets○

Path between source and receiver - enclosures for noisy machines, soundproof doors

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Climate

Four primary variables that define climate:

Air temperature1. Humidity - usually relative humidity2. Air movement3. Radiation from surrounding objects, including the sun4.

Most comfortable working environment (called the comfort zone):

Air temperature = 19 to 26°C (66 to 79°F)• Relative humidity = 50%• Slow air movement = 0.2 m/s (0.64 ft/sec)•

Heat Stress occurs when body absorbs more heat than it gives off, raising the body core temperature. Illnesses resulting from heat stress:

Heat rash - areas of skin erupt into red or white bumps due to inflammation of sweat glands1.

Heat cramps - spasms of muscles in physical labour2. Heat exhaustion - muscle weakness, nausea, dizziness3. Heatstroke - fever, dry skin, convulsions, coma (in extreme cases: death)4.

Reducing Heat Stress:

Administrative controls:

Provide water for workers○

Frequent rest breaks○

Limit times in hot environment○

Work in the shade, not in the sun○

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Engineering controls:

Provide air conditioning if feasible○

Provide fans○

Shield radiant heat sources○

Wear protective clothing○

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Cold Stress:

Drop in core body temperature below normal due to net heat loss from combination of low temperature and air movement

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Windchill factor

Estimates cooling effect of moving air on exposed skin○

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Can lead to

Frostbite - tissue freezes and ice crystals form in tissue cells.

Commonly occurs in hands and feet▪

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Hypothermia - when body core temperature is at or below 35°C (95°F)○

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Reducing Cold Stress:

Administrative controls:

Limit work times in cold environment○

Allow frequent breaks to warm hands○

Drink hot beverages○

-

Engineering controls:

Proper clothing (layering principle - multiple layers of clothing)○

Provide gloves○

Provide space heaters when building is cold○

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Refers to the size and shape of a facility as well as the relative locations and shapes of the functional areas (e.g., departments), equipment, workstations, storage spaces, aisles, and common areas (e.g., restrooms)

Concerned with problems of

Laying out a new facility○

Making changes in an existing facility○

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Also called plant layout

Usually associated with production plants○

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Fixed-position layout

Layout in which product remains in one location during fabrication, and workers and equipment are brought to the product:

Suited to low production quantities and high product variety• Reason for keeping product in one location:

Product is big and heavy○

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Typical plants: assembly and fabrication

Much manual labour○

Equipment is portable or mobile○

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Advantages Disadvantages

Very high mix and product flexibility• Product not moved or disturbed• High variety of tasks for staff•

Very high unit costs• Scheduling of space and activities can be difficult• Can mean much movement of equipment and staff•

Process (or functional) layout

Layout in which equipment is arranged according to function:

Facility Layout

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Suited to low and medium production quantities and medium to high product variety•

Different parts or products are processed through different operations in batches

Each batch follows its own routing○

Material handling activity is significant○

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Advantages Disadvantages

High mix and product flexibility• Relatively robust in the case of disruptions

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Relatively easy supervision of transforming resources

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Low facilities utilization• Can have very high work-in-progress or customer queuing

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Complex flow can be difficult to control•

Product (or line) Layout

Layout in which workstations and equipment are located along the line of flow of the work units:

Suited to high production quantities and low product variety• Work units typically moved by powered conveyor• At each workstation, a small amount of the total work content is accomplished on each work unit

Each station specializes in its task, thus achieving high efficiency○

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Advantages Disadvantages

Low unit costs for high volume• Gives opportunities for specialization of equipment•

Materials movement is convenient •

Can have low mix flexibility• Not very robust if there is disruption• Work can be very repetitive•

Cellular layout

Layout in which work units flow between stations, as in a production line, but each station can cope with a variety of part styles without the need for time-consuming changeovers:

Combination of product and process layouts

Tries to combine efficiency of product layout with versatility of process layout○

Neither objective is achieved perfectly, but it is more efficient than a process layout and more versatile than a product layout

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Based on principles of group technology•

Advantages Disadvantages

Gives a compromise between cost

- Can be costly to rearrange existing layout•

variety operations

Fast throughput• Potential good staff motivation•

Can give lower equipment utilization •

Combination layouts

Fixed-position and process layout:

Shipyard - ships made in modules

Parts fabricated in process layout○

Modules built in fixed-position layout○

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Off-site construction• Fixed-position and product layout:

Commercial airplanes (e.g., Boeing 747)

Fabrication begins with fuselage and proceeds through 7 or so stations where specialized workers assemble parts and modules to airplane

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Production flow analysis

Important factors in plant layout:

Q production quantity

number of units of a given part or product that the facility produces

P product variety number of different product designs or types made in the plant
Group parts into part families1. Identify processes2. Identify part families using the processes3. Develop a part-machine incidence matrix4. Perform clustering analysis to identify cells5.

Rank order clustering: Sort rows based on binary value (decreasing)1. Sort columns based on binary value2. Repeat process until no changes in sorting3. Use the clusters to create machine cells4.

DON'T FORGET TO RECALCULATE THE BINARY VALUES

Note that routing is not considered in this analysis

Typical problems with rank order clustering:

Presence of outliers and/or voids in the finished clusters

Solve by replicating machines or removing redundant machines○

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Clustering algorithms will only suggest facility reorganization based on existing routes• Need to get representative routes and ensure routes are accurate•

Levels of detail in plant layout: Site layout - shows building location on property1. Block layout - shows arrangement and sizes of departments in building2. Detailed layout - shows workstations and equipment in each department3. Workstation layout - indicates arrangement of equipment, workbench, WIP inventory, and worker floor space for each workstation type 4.

Objectives in layout planning:

Efficient movement of materials and people• Logical work flow and minimum travel distances• Efficient utilization of space• Safety and satisfaction of those who use the facility• Flexibility to meet changing future requirements• Advancing the operational mission of the facility•

Steps in systematic layout planning: Determine requirements and collect data1. Analyse material flows2. Define activity relationships and develop activity relationship chart3. Construct activity relationship diagram4. Determine space requirements5. Construct space relationship diagram6. Make adjustments and add allowances7. Develop block layout8. Develop detailed layout9.

Requirements and data

Starting specifications in plant layout design:

Specific product or set of products that will be produced• Particular set of manufacturing and/or assembly processes that will be performed• Specified quantities of parts and products that will be produced•

Rarely are specifications for two plants the same.

P Product What products and products

Q Quantity How many of each product

R Routing Manufacturing processes through which parts and products will be routed

S Supporting services Utilities, restrooms, offices, etc.

T Time issues How many shifts per week? By when must the building project be completed?

Analyse material flows

Possible charting techniques to analyse material flows:

Operation chart - shows sequence of processing, assembly, and inspection operations for products

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Flow process chart - shows processing steps and other details for parts production• From-To chart - shows quantities and directions of material flows between departments•

Systematic layout planning

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Part routing matrix - shows which operations each part is routed through during production

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Activity relationships

Indicate relative need to place activities or departments in close proximity to each other.

Defined by closeness ratings:

Closeness rating

Definition Numerical value

A Absolutely necessary for departments to be next to each other

4

E Especially important 3

I Important 2

O Ordinary 1

U Unimportant 0

X Undesirable to locate departments near each other

-5

Closeness rating factors: Material flow - most important, sometimes the only factor1. Need

for contact between personnel2. Use of same equipment3. Sharing of common records4.

Sharing of supervision or technical staff5. Use of same utilities6. High noise level7. Emission of fumes, odours8.

From-to chart Flow-between chart

Activity relationship chart:

Activity relationship diagram

Graphical means of displaying the closeness ratings among pairs of activities (departments)

Uses nodes to represent activities• Nodes are connected by lines indicating the closeness ratings• Closeness ratings are identified by

Colour○

Different types of lines○

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Closeness rating

Line type

A ///

E ///

I //

O /

U (no line)

X \\\

Analyst attempts to rearrange nodes so that nodes with high closeness ratings are located near each other.

Space requirements

Objective is to determine area requirements for each activity (department). Procedure:

List all workstation types

1.

Determine floor space requirement for each station type

2.

Determine number of stations of each type:

3.

Where:

Total workload that must be accomplished in a certain period, expressed in hrs of work per period

Available time at the workstation in the same period.

If workload for station i consists of multiple parts or products:

4.

Where:

Quantity of part type j

Cycle time of part type j

Determine total area required for each workstation type:

5.

Determine total area in each department by summing areas for all station types in that department:

6.

Where:

Area of department k and summation is carried out over all workstation types in department k

Four types of workstations will be included in department 1. The floor area for each type, workloads per week and available times per week are given in the table below. Determine the total area required for these four workstation types.

Workstation type

Area per workstation (m²)

Workload per week (hr) Available time per week (hr)

1 16 262 38

2 20 123 36

3 15 244 38

4 25 89 37

1 16 262 38 6.89 7 122

2 20 123 36 3.42 4 80

3 15 244 38 4.84 5 75

4 25 89 37 2.80 3 72

Total 349 m²

Space relationship diagram

Extension of activity relationship diagram in which nodes representing departments are now assigned areas that are proportional to areas calculated in previous step

Shapes of nodes may be changed to fit the nodes into the diagram• But relative positions of blocks remain as in the last iteration of the activity relationship diagram

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Assume that the total areas have been determined for the six departments in the previous example. The total area values for each department are listed in the table below. Construct the space relationship diagram for this data.

Department (k) 1 2 3 4 5 6

Total area (DA_k) 349 147 445 98 255 196

Adjustments and allowances

Personnel requirements:

Rest rooms• Locker rooms,• Food services• Plant entrances and exits•

Material handling methods:

May affect floor space and building height• Storage facilities:

Treat as separate department or add allowance• Aisle space:

Add a percentage allowance• Offices:

For individual departments• Building features:

Walls• Column locations•

Site conditions:

Parking• Landscaping•

Limitations:

Budget - construction costs may impose limitation on building size• Building codes - may affect location of building on property as well as structural and utility details

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Safety requirements - some included in building codes• Existing building - if layout is planned around an existing building, then shape and size of building is a limitation on plant layout design

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Develop block layout

Develop alternative layout plans based on space relationship diagram:

Different shapes and aspect ratios• Evaluate alternatives Select the best

Compromise between competing objectives, modifying considerations, and practical limitations

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Space relationship diagram Rectangular building shape

Numerical scoring methods to evaluate the block layout:

Adjacency score• Layout efficiency rating•

Adjacency score is computed by summing the numerical closeness rating values for all contacting pairs of departments:

Where:

Adjacency variable

1 if common border• 0 if no common border•

Numerical value of closeness ratings

A = 4• E = 3• I = 2• O = 1• U = 0• X = -5•

Layout efficiency rating:

Where:

AS Adjacency score

And:

Where:

Determine the adjacency score and layout efficiency rating for the block layout:

Develop detailed layout

Filling in the block layout with the details of how each department will be arranged:

Locations and areas of workstations and equipment, aisles, department offices, storage areas, etc.

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Layout design provides a check on the original calculations

If details do not fit into spaces correctly, then there may be an error in original space calculations

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Lean production is a management philosophy that aims to optimize business processes by eliminating waste in all aspects of production, from design to distribution, while maintaining high quality, ultimately focusing on creating maximum value for the customer with minimal unnecessary steps and resources.

House of quality/Toyota Production System

Key points about lean production:

Focus on value:

Identifying what truly adds value to the product from the customer's perspective and focusing efforts on those activities.

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Waste reduction:

Actively seeking to eliminate all forms of waste including excess inventory, unnecessary steps in the production process, defects, and waiting time.

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Continuous improvement:

A culture of constant evaluation and refinement of processes to achieve greater efficiency.

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Just-in-time (JIT) inventory:

Ordering materials only as needed to minimize inventory holding costs.

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Pull system:

Production is driven by customer demand, meaning goods are only produced when needed.

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Core principles of lean production:

Define value: Identify what the customer truly values in the product. • Identify the value stream: Analyse all steps in the production process to distinguish value-adding from non-value-adding activities.

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Create flow: Ensure smooth and continuous movement of materials through the production process.

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Establish a pull system: Produce goods based on actual customer demand rather than forecasts.

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Strive for perfection: Continuously seek to improve processes and eliminate all waste. •

Lean production

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Taiichi Ohno's seven forms of waste (Muda): T I M W O O D S

Transportation: Unnecessary conveyance of products, from one location to another, or handoff from one employee to another Caused by:

Poor layouta. Complex material handlingb. Large batch sizesc. Multiple storage locationsd. 1.

Inventory: More stock and work in progress than is required to serve customer requirements. Caused by:

Large batch sizesa. Unbalanced work flows, operations working at different ratesb. Long changeover timesc. Producing "just in case"d. 2.

Motion: Unnecessary physical movements – work smarter not harder! Caused by: Poor workstation layouta. Poor design of production methodb. Reorientation of materialsc. Large batch sizesd. 3.

Waiting: Idle time, waiting to produce added value work. Caused by: Processes not synchroniseda. Waiting for other resources:

Crane or forklifti. Inspectionii. Machine setteriii. Machine to complete an automatic cycleiv. b. 4.

Overproduction: Producing more than the customer requires or producing earlier than is required. Caused by

Large batch sizesa. Poor planning, working to forecastb. Unreliable processesc. Poor synchronisationd. 5.

Overprocessing: Unnecessary work that does not contribute value to the product. Caused by: Poor understanding of customers requirementa. Poor designsb. Inadequate quality specificationsc. Poor standardisation of techniquesd. 6.

Defects: Defects cause additional work to correct, or even worse customer dissatisfaction. Caused by:

Incapable machinesa. Poor skills and poor trainingb. Operator errorsc. Poorly designed and inadequately specified proceduresd. 7.

Only the first 7 wastes were identified by Taiichi Ohno Skill: Untapped knowledge and human potential. Caused by:

Not involving workers in process design and problem solvinga. A mindset that “managers think, workers do”b. Not formalising and recording ideas and improvements c.

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Elimination of waste

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Production and delivery of exactly the required number of each component to the downstream operation in the manufacturing sequence just at the moment when the component is needed. Minimizes:

Work-in-process • Manufacturing lead time •

Pull

In manufacturing, "push" refers to producing goods based on forecasted demand, essentially pushing products out to the market before a customer order is placed, while "pull" means producing goods only when there is actual customer demand, effectively pulling products through the production line as needed; essentially, push relies on prediction while pull reacts to real-time orders, with pull generally considered a more efficient approach in terms of inventory management and reducing waste. Key differences:

Production Trigger:

Push systems produce based on sales forecasts, while pull systems produce based on actual customer orders.

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Inventory Levels:

Push systems tend to have higher inventory levels due to potential overproduction based on inaccurate forecasts, whereas pull systems aim to minimize inventory by only producing what is needed.

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Flexibility:

Pull systems are generally more flexible in responding to changing market demands compared to push systems.

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Push A clothing manufacturer produces a large batch of winter jackets in the summer based on anticipated demand for the upcoming season.

Pull A custom furniture maker only starts building a piece of furniture once a customer places an order with specific design specifications.

Pull the product from the system at the rate of customer demand. Wherever practicable, pull the product through the whole system at the same rate. Reduced inventory costs, improved responsiveness to customer demand, and less waste due to overproduction. But because of technical constraints it may be necessary to push product to a certain point. Identifying the push-pull decoupling point is key.

Kanban system

Kanban is a visual workflow management method used to optimise efficiency in teams, particularly in software development, manufacturing, and project management. It originates from the Toyota Production System and is designed to help teams balance work demand with available capacity. Toyota's way of implementing a pull system of production control.

Kanban means "card" in Japanese.

Just-in-time production

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Visualise the work – Tasks are represented as cards on a board, typically divided into columns (e.g., "To Do," "In Progress," "Done"). This makes the flow of work transparent and easy to manage.

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Limit work in progress (WIP) – By setting limits on the number of tasks in progress, Kanban prevents bottlenecks and overloading team members.

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Manage flow – Teams focus on improving the movement of work through different stages rather than pushing more tasks into the system.

-

Make process policies explicit – Everyone involved should understand the workflow rules and criteria for moving tasks between columns.

-

Implement feedback loops – Regular reviews (such as daily stand-ups or retrospectives) help improve efficiency and adapt to changes.

-

Improve collaboratively – Kanban encourages continuous improvement based on data and team discussions.

-

A Kanban board can be physical (e.g., a whiteboard with sticky notes) or digital (e.g., Trello, Jira, Asana). It typically consists of:

Backlog (optional) – A list of tasks waiting to be prioritised. • To Do – Work ready to be started. • In Progress – Tasks currently being worked on. • Review/Testing (optional) – Tasks awaiting approval or quality checks. • Done – Completed tasks. •

Benefits of Kanban:

Increases efficiency by reducing bottlenecks. • Provides transparency and clarity on team workload. • Encourages continuous improvement. • Adaptable to various industries and teams. •

The two main types of Kanban are Production Kanban and Transport Kanban. They serve different purposes in workflow management, particularly in manufacturing and lean systems.

Production Kanban This type of Kanban signals when a process should produce more items. It is used to regulate production based on demand, ensuring that new work is only started when there is a need for it.

Purpose: Controls the amount of work being produced.▪

Example: A Kanban card in a factory instructs a workstation to produce a specific quantity of a part when stock reaches a certain level.

▪

Application: Used in Just-in-Time (JIT) manufacturing to prevent overproduction.

▪

1.

Transport Kanban This type of Kanban signals when materials or products should be moved from one location to another. It ensures that components or finished goods are transported only when required, reducing unnecessary stockpiling.

Purpose: Regulates the movement of materials within a system.▪

Example: A warehouse Kanban system where a signal card tells a worker to transport components to the assembly line just in time for production.

▪

Application: Used in logistics, warehouses, and supply chain management to optimise inventory flow.

▪

2.

Feature Production Kanban Transport Kanban

Function Controls production Controls movement

Trigger When more units are needed When stock needs replenishing

Goal Prevents overproduction Prevents unnecessary transport

Example A signal to manufacture more car parts

A signal to move parts to an assembly line

Reducing set-up times

Single Minute Exchange of Dies

Analyse the set-up process1. Classify setup operations2. A→B

Waste - Operation which do not add value to the setupa. **Internal Setups** - Operations that can only be performed while the machine is shut down

b.

External Setups - Operations that can be performed without shutting down the machine

c.

Eliminate waste3. Convert as many internal setups as possible to external setups 4. B→C

Improve internal setups to reduce time5. C→D

quick change tooling / connectionsa. standard base platesb. quick locate and fit connectionsc.

Improve external setups6. C→D

Use checklist to avoid omission.a. Use specially designed cart to help organize toolsb.

Organize workplace to reduce searchc.

Develop standard operating procedure7. Evaluate the performance of the new operation and continuously improve8.

External work elements:

Can be accomplished while previous job is still running• **Strategy:** Design the setup tooling and plan the changeover procedure to permit as much of the setup as possible to consist of external elements

•

Examples:

Retrieve tooling for next job from tool crib○

Assemble tools for next job○

Reprogram machine for next job○

-

Internal work elements:

Use time & motion study and methods improvement to minimize the sum of the internal work element times

-

Use two workers rather than one• Eliminate adjustments in the setup• Use quick-acting fasteners rather than bolts and nuts• Use U-shaped washers instead of O-shaped washers• Use of interrupted screw•

Other requirements for JIT

This involves distributing changes in product mix and quantity as evenly as possible over time. Instead of producing large batches of a single product, production is balanced to meet demand with minimal fluctuations. This helps reduce bottlenecks, inventory buildup, and sudden strain on resources.

On-time delivery of components JIT requires that raw materials and components arrive exactly when needed, neither too early (causing excess inventory) nor too late (causing delays). This depends on precise scheduling and a reliable logistics system.

Defect-free components and materials Since JIT operates with minimal inventory, there is little room for defective parts. If defects occur, they can halt production immediately.

Suppliers must provide high-quality materials, and quality control should be integrated into every step of the process.

Reliable production equipment Equipment breakdowns can disrupt JIT operations because there is no excess stock to cushion delays. Regular preventive maintenance and fast-response repair systems are critical to keeping production running smoothly.

Workforce that is cooperative, committed, and cross-trained Employees should be adaptable and skilled in multiple roles. This flexibility allows workers to step in where needed, helping to maintain flow even if there are temporary shortages or equipment issues. A strong culture of teamwork and problem-solving is also essential.

Dependable supplier base Since JIT relies on frequent and small deliveries, suppliers must be highly reliable. Any delays or inconsistencies can cause major disruptions. Close collaboration, good communication, and long-term partnerships with suppliers are key to ensuring a steady and timely supply of materials.

“Automation with a human touch” Production machines operate autonomously as long as they are functioning properly. When they do not function properly (e.g., they produce a defect), they are designed to stop. Key automation topics:

Stop the process (Jidoka)• Error prevention (Poka-yoke)•

Jidoka

Automation with a Human Touch Jidoka is one of the two pillars of the Toyota Production System (TPS) (the other being Just-in-Time). It focuses on built-in quality by ensuring that defects are detected and corrected as soon as they occur. The word "Jidoka" translates to "autonomation", meaning automation with intelligence, where machines or workers can stop production when an abnormality is found. The principle of zero-defects:

Key Principles of Jidoka:

Detect an abnormality – A problem is recognised through sensors, workers, or inspection points in the process.

-

Stop production immediately – The affected machine, workstation, or production line stops to prevent the defect from moving forward.

-

Fix the problem at the source – The root cause of the issue is investigated and corrected before production resumes.

-

Implement countermeasures – A long-term solution is put in place to prevent the same issue from happening again.

-

Why is Jidoka Important?

Ensures product quality – Defects are caught early rather than after production. • Prevents waste – Reduces defective products that would otherwise need rework or scrapping.

-

Enhances efficiency – Production doesn't continue with faulty components, saving time and resources.

-

Encourages worker involvement – Workers actively contribute to quality improvement by stopping and fixing defects.

-

Examples of Jidoka in Action

Andon System in Car Manufacturing

In Toyota factories, workers have the authority to pull an Andon cord (a visual signal system) if they notice an issue.

-

This immediately halts production and alerts supervisors to fix the problem before continuing.

-

The goal is to address defects at the source rather than letting faulty cars reach customers.

-
-

Automated Quality Inspection in Food Production

-

Autonomation

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Many food factories use vision sensors on conveyor belts to detect misshaped or contaminated products.

-

If a defective product is found, the system automatically stops or diverts the bad item, ensuring only quality food reaches consumers.

-

Printing Industry – Paper Jam Detection

In a printing press, Jidoka is applied through sensors that detect a paper jam.○

Instead of continuing and damaging the printer, the machine stops itself and alerts an operator to fix the issue before resuming.

-

-

Poka-yoke

Mistake-Proofing Poka-yoke is a Japanese term meaning "mistake-proofing" or "error-proofing." It is a technique used to eliminate mistakes before they happen by designing systems, processes, or devices that either prevent human errors or detect them immediately.

Key Principles of Poka-yoke:

Prevention-Based Poka-yoke

Prevents errors from happening in the first place.○

-

Detection-Based Poka-yoke

Identifies errors when they occur and alerts the user.○

-

Why is Poka-yoke important? Reduces human errors – Mistakes are prevented, ensuring higher quality and fewer defects. Improves safety – Reduces accidents caused by incorrect procedures. Enhances efficiency – Saves time and resources by avoiding errors in production and daily tasks.

Examples of Poka-yoke in Action

USB Plug – One-Way Insertion (Prevention-Based)

USB connectors are designed to be inserted only one way.○

This prevents users from inserting them incorrectly, avoiding damage or malfunction.

-
-

Microwave Safety Sensor (Detection-Based)

Microwaves do not operate unless the door is properly shut.○

This prevents users from running the microwave with an open door, ensuring safety.

-
-

Car Seatbelt Warning System (Detection-Based)

Modern cars use a seatbelt sensor that alerts drivers with a beeping sound if they try to drive without buckling up.

-

This prevents safety hazards by reminding the driver to fasten the seatbelt.○

-

Manufacturing – Colour-Coded Parts (Prevention-Based)

In electronics factories, similar-looking components are colour-coded or have different connector shapes.

-

This prevents assembly mistakes, ensuring workers place the right parts in the right places.

-
-

Medication Dispensers in Hospitals (Prevention-Based)

Some hospitals use electronic dispensers that only allow the correct medication to be taken at the right time.

-

This prevents medical errors by ensuring patients receive the correct dosage.○

-

Levels of intervention:

Level Indicator Signal Control Guarantee

Warning Passive warning Active warning Active intervention Avoid mistakes

Responsibility Identify and react correctly

React correctly Guide to correct behaviour Deviation possible

Only correct behaviour possible

Fail safe power LOW MODERATE STRONG ABSOLUTE

Key Differences Between Jidoka and Poka-yoke:

Feature Jidoka – "Automation with a Human Touch" Poka-yoke – "Mistake-Proofing"

Purpose Detects and stops defects early Prevents mistakes from happening

Action Halts production to fix problems Eliminates or alerts errors automatically

Main Focus

Quality control and defect detection Human error prevention

Example Andon cord stopping production when a defect is found

A plug that only fits in one direction

Although Jidoka and Poka-yoke are different, they complement each other in improving quality and reducing waste.

Jidoka ensures quality is maintained throughout the process by detecting defects early and stopping production if needed.

-

Poka-yoke prevents mistakes from occurring in the first place, reducing the need for Jidoka interventions.

-

For example, in car manufacturing:

A Poka-yoke system might use a sensor to ensure bolts are tightened correctly before moving to the next station.

-

If an issue is detected, Jidoka stops production and triggers an alert so the problem can be fixed immediately.

-

Both Jidoka and Poka-yoke are essential for achieving high-quality, defect-free production in manufacturing and everyday processes. Jidoka ensures that defects do not continue down the production line by detecting and stopping them early. Poka-yoke eliminates human errors and process mistakes before they happen. By combining both, companies can enhance efficiency, reduce waste, and improve product quality, leading to a more reliable and cost-effective production system.

Continuous improvement

Kaizen is a Japanese philosophy that focuses on continuous, incremental improvements in all aspects of a business or personal workflow. The word "Kaizen" translates to "change for better" and is widely used in Lean manufacturing, business management, and personal development.

Key Principles of Kaizen:

Continuous Improvement – Small, ongoing positive changes lead to major improvements over time.

-

Eliminate Waste – Focus on efficiency by reducing unnecessary steps, time, or resources.

-

Empower Employees – Encourage workers at all levels to suggest and implement improvements.

-

Standardise Best Practices – Once an improvement is successful, make it a standard procedure.

-

Focus on Process, Not Just Results – Improving the way things are done leads to better long-term success rather than quick fixes.

-

Why is Kaizen Important?

Boosts productivity – Small refinements make processes smoother and faster. • Reduces waste and costs – Eliminating inefficiencies saves time, money, and resources. • Enhances quality – Continuous monitoring leads to fewer defects and errors. • Encourages teamwork and engagement – Employees feel valued when they contribute ideas for improvement.

-

Examples of Kaizen in Action:

Toyota Production System (TPS) – Toyota workers regularly suggest small changes to improve assembly line efficiency and product quality.

-

Retail Checkout Process – Supermarkets use Kaizen by tweaking cashier workflows, reducing unnecessary movements to speed up checkout times.

-

Office Productivity – Employees in an office setting implement 5S principles (Sort, Set in order, Shine, Standardise, Sustain) to create a more organised and efficient workspace.

-

Visual workplace

Visual Management is a technique used in Lean to make information clear, visible, and easy to understand at a glance. It helps teams quickly identify issues, track progress, and ensure efficiency by using visual cues instead of relying on complex instructions.

Key Goals of Visual Management:

Increase Transparency – Everyone in the organisation can see the current state of work. •

Identify Issues Immediately – Problems are visible, allowing for quick intervention. • Improve

Efficiency – Employees spend less time searching for information. • Standardise

Communication – Ensures consistency in processes and expectations. •

Examples of Visual Management in Action:

Kanban Boards – Used in project management to track tasks (To-Do, In Progress, Done).

-

Andon Lights in Manufacturing – Green, yellow, and red lights indicate machine or

-

Worker involvement

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Shadow Boards for Tools – Outlines of tools on a board show where each tool belongs, making it easy to spot missing items.

-

Colour-Coded Floor Markings – Used in warehouses or hospitals to guide movement, ensuring safety and efficiency.

-

Dashboards for Performance Metrics – Factories or offices display production rates, defects, or sales figures in real time.

-

Why is Visual Management Important? Reduces errors by providing clear instructions.1.

Improves response time by making problems obvious.2. Supports teamwork by ensuring everyone understands the process.3.

5S is a structured approach to workplace organisation that ensures efficiency, cleanliness, and safety. It consists of five Japanese principles, each starting with 'S'.

The 5 Steps of 5S:

Sort (Seiri) – Remove unnecessary items from the workspace

Clearing out old tools or paperwork that are no longer needed.

1.

Straighten (Seiton) – Arrange essential items so they are easy to access. Organising tools in a fixed location for quick retrieval.

2.

Shine (Seiso) – Keep the workspace clean and tidy. Regularly cleaning machines to prevent breakdowns.

3.

Standardise (Seiketsu) – Establish consistent procedures for maintaining order. Creating a checklist for daily cleaning and tool storage.

4.

Sustain (Shitsuke) – Make 5S a habit by continuously improving and training employees. Conducting regular audits to ensure 5S practices are followed.

5.

Why is 5S Important?

Increases efficiency – Everything is where it should be, reducing wasted time. • Enhances safety – A well-organised workspace reduces risks of accidents. • Boosts morale – Employees work better in a clean, structured environment. • Supports continuous improvement (Kaizen) – Helps maintain high standards over time. •

Examples of 5S in Action:

Manufacturing Tools and machines are kept in designated areas, reducing downtime.

Office Workspaces

Files and documents are arranged systematically for easy retrieval.

Hospitals Medical equipment is colour-coded and stored in labelled drawers for quick access.

Restaurants Kitchens follow strict cleaning and organisation practices to maintain hygiene and efficiency.

Total productive maintenance Goal: zero breakdowns TPM = integration of preventive and predictive maintenance to avoid emergency maintenance It aims to achieve 100% on-demand equipment availability by eliminating:

Equipment Breakdowns and other unplanned downtime • Scrap and rework by poor equipment performance • Reduced productivity due to running at reduced speeds, idling or stoppages requiring operator attention

•

Equipment startup losses •

Maintenance Outcome

Emergency Repair equipment that breaks down

Preventive Routine repairs to avoid breakdowns

Predictive Anticipating malfunctions before they occur

Operators do preventive maintenance and maintenance department must be just a back-up.

Overall Equipment Effectiveness

AR Availability rate

PR Performance rate

QR Quality rate

Availability Rate (AR) in % measures how long equipment is not producing parts due to downtime:

Performance Rate (PR) measures actual machine output as a % of standard machine output:

Quality Rate (QR) measures quality in terms of % of good product:

But this definition of Overall Equipment Effectiveness (OEE) is not useful!

Why do companies fail to “copy” TPS?

Well over 50% companies fail at lean manufacturing • Change management and gaining people commitment are the key barriers • Need to put the entire system in place, and not just bits and pieces of it • Overlooked cultural factors in Toyota

The presence of guaranteed employment in Toyota ○

The ability to make mistakes ○

-

Toyota avoids bad decisions from top management, because management themselves understand the issue first-hand

-

Need to adapt to the organisational requirements and culture•

What is maintenance?

classical view “... fixing broken items.”

contemporary

view

“...all activities aimed at keeping an item in, or restoring it to, the physical state considered necessary for the fulfilment of its production function.”

Inspection• Cleaning• Parts replacement• Repairs• Overhauls• Upgrade•

Key decisions in maintenance:

What maintenance action to be taken, and when?

Replace○

Repair○

Overhaul○

-

When to replace components?• How often to inspect, and what technique is to be used?• How many spare parts are to be kept in stock?• When to retire/replace an equipment?

Like-for-like○

New○

-

Why is maintenance important? RELiCS

Reliability○

Efficiency○

Life extension○

Compliance○

"Shine"○

Maintenance strategies

Corrective maintenance:

“Run-to-failure” or “reactive” maintenance•

Approaches to maintenance

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If it ain't broke, don't fix it – sounds simple and reasonable• No money spent on maintenance until the machine stops working• Disadvantages:

High spare parts inventory○

High overtime costs○

Long machine downtime○

Low production availability○

Spare machines required○

Knock-on effects on other machines and overall loss of production○

Danger of unsafe failures○

-

Planned maintenance:

Maintenance is time-driven or age/usage driven• Based on elapsed time/hours of operation/
number of parts produced• Time between maintenance based on statistical data or
manufacturer's recommendations

-

Scheduled maintenance costs are much lower than corrective maintenance costs• Easier to
manage• Treats all machines as same•

Condition-based maintenance:

Regular monitoring of actual mechanical condition of the machine and other indicators of
operating condition provide data for the optimal time to act.

-

Regular monitoring of actual mechanical condition of the machine and other indicators of
operating condition provide data for the optimal time to act.

Measured values of sensors○

Condition Monitoring○

State of components○

Prediction of remaining lifetime○

Predicted time of failure○

Computerised Maintenance Management○

-

Costs

Design of monitoring system○

Monitoring equipment (sensors, analysis tools, etc.)○

Staff training○

Labour costs for inspection, measurement, analysis, etc.○

-

Savings

Elimination/reduction of unexpected breakdowns○

Increased time between repairs○

Reduction of spare part stock○

Reduction in insurance premium○

-

Component replacement decision on the basis of operational efficiency improvement:

Component replacement interval

Operating cost at time t

Replacement cost

Total cost per unit time = total cost in interval $(0, t_r)$ / length of interval

Total cost in interval = cumulative operating cost + replacement cost

The trend in operating cost for an item is of the form:

Optimising maintenance decisions

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Where: $A = £100$ $B = £80$ $k = 0.21/\text{week}$

C_r , the cost of replacement is £100. Calculate the optimal replacement policy.

Calculate $C(t_r)$ for $t_r = 1, 2, 3, \dots$

t_r 1 2 3 4 5 6 7

$C(t_r)$ 127.8 84.7 74.0 70.9 70.5 71.5 72.5

Preventive replacement

Block policy Age-based policy

C_p = preventive replacement cost C_f = failure replacement cost f_t = pdf of failure times t_p = replacement age $C(t_p)$ = Total expected replacement cost per unit time To determine optimal replacement age of the item to minimise the total expected replacement cost per unit time.

Shaded area: probability of preventative cycle

White area: probability of failure cycle

Total expected replacement cost per cycle = cost of a prev. replacement $\times$ prob. of prev. cycle + cost of failure $\times$ prob. of failure cycle

Expected cycle length = length of a preventive cycle (t_p) $\times$ probability of the preventive cycle ($R(t_p)$) + expected length of a failure cycle ($M(t)$) $\times$ probability of a failure cycle ($1 - R(t_p)$)

The truncated failure time distribution is given by:

The mean time to failure of the complete distribution is:

If a preventive replacement occurs at time t_p then, the mean time to failure is the mean of the unshaded portion of the figure above because the shaded area is an impossible region for failures. The mean of the unshaded area is given by:

Hence:

Need to minimise $C(t_p)$

1000 failures of a particular component in a machine tool were analysed:

Age (in hrs) 0-10 '000 10-20 '000 20-30 '000 30-40 '000

Number of failures

250 250 250 250

Cost of failure replacement = £200 Cost of preventive replacement = £100 What is the optimal preventive replacement age in hours? Consider values of $t_p = 10000, 20000, 30000, 40000$ hours.

Prev. replacement time

Failure freq.

Prob of failure before t

Prob of survival until t

Exp. time to failure before t

Expected Cost

Expected Cycle Length

Total Cost/1000 hours

t $f(t)$ $F(t)$ $R(t)$ $M(t)$ 200 $\times F(t)$ + 100 $\times R(t)$

$M(t) \times F(t) + t \times R(t)$

10000 0.25 0.25 0.75 5000 125 8750 14.3

20000 0.25 0.5 0.5 10000 150 15000 10

30000 0.25 0.75 0.25 15000 175 18750 9.3

40000 0.25 1 0 20000 200 20000 10

Replace after 30000 hours.

t f(t) F(t) R(t) M(t) Expected Cost

Expected Cycle Length

Total Cost/1000 hours

10000 0.125 0.125 0.875 5000 112.5 9375 12.0

20000 0.375 0.5 0.5 12500 150 16250 9.2

30000 0.375 0.875 0.125 17857 187.5 19375 9.7

40000 0.125 1 0 20000 200 20000 10

Replace after 20000 hrs (Actual optimal solution is 22425 hrs)

Dobson's table for age-based policy

Assuming a Weibull lifetime distribution with shape and scale parameters β and η respectively:

1000 failures of a particular component in a machine tool were analysed and the times to failure was fit with a Weibull distribution with the following parameters:

$\eta = 30000$ hrs $\beta = 2.5$ Cost of failure replacement = £20000 Cost of preventive replacement = £100

What is the optimal preventive replacement age in hours?

Manufacturing strategy is a pattern of decisions, both structural and infrastructural, which determine the capability of a manufacturing system and specify how it will operate in order to meet a set of manufacturing objectives which have been derived from business objectives.

Work is defined as an activity in which a person exerts physical and/or mental effort to accomplish a given task or perform a duty.

Task Work elements

Basic motion elements

Method study is the systematic recording and critical examination of existing and proposed ways of doing work, as a means of developing and applying easier and more effective methods and reducing costs. (BS 3138:1979). Other terms you may come across for methods study:

Methods engineering• Work simplification• Process re-engineering• Business process re-engineering•

Typical objectives of method study:

Increase productivity and efficiency• Reduce cycle time• Reduce product cost• Reduce labour content• Improve customer satisfaction• Improve product and/or service quality• Reduce lead times and improve workflow• Increase flexibility of work system• Improve worker safety• Enhance the environment (both inside and outside the factory)•

Work

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The basic procedure for method study is as follows:

SELECT the work to be studied

RECORD all the relevant facts about the present method by direct observation

EXAMINE those facts critically and in ordered sequence

DEVELOP the most practical, economic and effective method

DEFINE the new method so that it can always be identified

INSTALL that method as standard practice

MAINTAIN that standard practice by regular checks

The selection of work to be studied can arise from many sources. Obvious areas are bottlenecks-where a particular operation is restricting output, labour intensive operations, operations involving much material handling, operations with high scrap or wastage, etc. A pareto analysis of operations cost or labour requirement can help in the selection. Other criteria:

Movement over long distances• Repetitive operations• Work with H&S concerns etc. •
Select

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Diagrams, charts and pictorial methods are widely used to:
improve understanding• facilitate communication •

Operation Charts

Graphical and symbolic representation of the operations used to produce a product. Two types of operations:

Processing and assembly operations Changing the shape, properties or surface of a material or workpart○

Joining two or more parts to form an assembly○

1.

Inspection operations Checking the material, workpart, or assembly for quality or quantity○

2.

Process maps

Process mapping techniques have been widely used to provide a detailed picture of the process or system of interest. Commonly used symbols for basic process maps:

Beginning/ending point of the process

Record

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Task or activity step

Decision point

Swim-lane process map:

Graphical and symbolic representation of the processing activities performed on something or by somebody. Principal types of process charts:

Material (or Flow) process chart - analysis of material or workpiece flow• Worker process chart - analysis of a worker performing a task•

Flow process chart

Uses five symbols to detail the work performed on a material or workpiece as it is processed through a sequence of operations and activities.

Operation Processing of a material

Inspection Check for quality or quantity

Move Transport of material to new location

Delay Material waiting to be processed or moved

Storage Material kept in protected location

Two-handed process chart

This is a derivative of the process chart and is used for the detailed analysis of an operator at a work station. The movements of both hands are charted individually and to a greater level of detail than on a normal process chart. The meaning of the conventional symbols is slightly modified. "Inspection" is generally omitted as it is of the same form as "Operation".

operation - used for grasp, use, release etc. of tool, component or material. • transport - used for the movement of the hand to or from the work, tool etc. • delay - used when the hand is idle • storage - redesignated "hold" used when the hand is used as a holding fixture •

This led on to micromotion study and then to predetermined time motion systems such as MTM, that we will look at later in the course.

Activity charts

A listing of the activities of one or more subjects (e.g., workers, machines) plotted against a time scale to indicate graphically how much time is spent on each activity. Types of activity charts:

Right-hand/left-hand activity chart (a.k.a. workplace activity chart)• Worker-machine activity chart• Worker-multimachine activity chart• Gang activity chart (a.k.a. multiworker activity chart)•

Multiple activity chart

A Multiple Activity Chart is a chart on which the activities of more than one subject (worker, machine, or item of equipment) are each recorded on a common time scale to show their inter-relationship. The activities of each subject are recorded on a separate bar or column against the time scale. Multiple activity charts are particularly useful for analysing and improving operations where there are a number of interrelated activities involving several operators and pieces of equipment. In very short-cycle operations where an operator is interfacing with one or more mechanical devices, the use of a multiple activity chart is almost an essential analysis technique.

Layout charts and string diagrams

These are used when looking at the layout of plant and equipment. They differ from flow process charts by showing movements plotted on a scale drawing. String diagrams, while appearing crude, provide an excellent way of getting to grips with a complex layout.

The activities recorded on the flow process chart fall into two main categories, namely: those in which something is actually happening to the material under consideration, i.e. it is being worked upon, moved or examined;

•

those in which the material is not being touched, i.e., delay or storage• The Activities in the first category may be further subdivided into three groups:

“MAKE READY” activities required to prepare the material or work and set it in position ready to be worked on.

•

“DO” operations in which some transformation is made to the material• “PUT AWAY” activities during which the work is moved from the workplace, this may include inspection.

•

Clearly the aim must be to maximise the ratio of "DO" operations as these are the only things that add value to the product, and ultimately provide the basis for the business.

All activities are subjected to a systematic questioning technique as follows:

the PURPOSE for which the activity is undertaken• the PLACE at which the activity is undertaken• the SEQUENCE in which the activity is undertaken• the PERSON by whom the activity is undertaken• the MEANS by which the activity is undertaken•

Examine with a view to ELIMINATING, COMBINING, REARRANGING or SIMPLIFYING those activities.

Query PURPOSE with the initial view of elimination

What is actually done, what is achieved?• Why is the activity necessary at all?•

Query PLACE, SEQUENCE and PERSON with the view of combining or rearranging:

Where is it being done? Why is it done at that particular place?• When is it done? Why is it done at that particular time?• Who is doing it? Why is it done by that particular person?•

Query MEANS with a view to simplifying How is it being done?

Why is it being done in that particular way? •

A second stage of questioning then inquires: What else might be done?

What should be done?• Where else might it be done?• Where should it be done?• When might it be done?• When should it be done?• Who else might do it ?• Who should do it ?• How else might it be done?• How should it be done?•

Examine

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The Principles of Motion Economy – first developed by Frank Gilbreth, expanded by others e.g., Ralph Barnes

Use of the Human Body• Design of the Workplace• Design of Tools and Equipment.•

Use of human body

Basic principles

Both hands should be fully utilised1. The two hands should begin and complete their movements at the same time.2. Motions of the arms should be symmetrical and in opposite directions and should be made simultaneously.

3.

The work should be designed to emphasise the worker's preferred hand.4. The two hands should not be idle at the same time except during periods of rest.5.

Rectangular bin

Bin with tray

Right hand

Left hand Both hands

Right hand

Left hand Both hands

Time to perform

1.04s 1.10s 1.48s 0.81s 0.91s 1.07s

The next five principles attempt to utilise the laws of physics to assist in the use of hands and arms while working.

Continuous curved movements are to be preferred to straight-line motions involving sudden and sharp changes in direction.

6.

Momentum should be employed to help the worker, but should be reduced to a minimum whenever it has to be overcome by muscular effort.

7.

The method should take advantage of gravity instead of opposing it.8. Rhythm is essential to the smooth and automatic performance of a repetitive operation. The work should be arranged to permit easy and natural rhythm whenever possible.

9.

Hand and body motions should be made at the lowest classification at which it is possible to do the work satisfactorily.

10.

Classifications of hand and arm motions:

Classification Defined as

1 Finger motions only

2 Finger and wrist motions

3 Finger, wrist and forearm motions

4 Finger, wrist, forearm, and upper arm motions

5 Finger, wrist, forearm, upper arm, and shoulder motions

The two remaining principles are recommendations for using body members other than the hands and arms:

The method should be designed to utilise the worker's feet and legs when appropriate.11. Work should be arranged so that eye movements are confined to a comfortable area, without the need for frequent changes of focus.

12.

Arrangement of workplace

Develop

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Principles: Definite and fixed stations should be provided for all tools and materials to permit habit formation.

1.

Tools and materials should be pre-positioned to reduce searching.2. Gravity feed, bins and containers should be used to deliver the materials as close to the point of use as possible.

3.

Tools, materials, and controls should be located within the maximum working area and as near to the worker as possible.

4.

Materials and tools should be arranged to permit the best sequence of motion. 5.

Provision should be made for adequate lighting, and the colour of the workplace should contrast with that of the work and thus reduce eye fatigue.

A chair of the type and height to permit good posture should be provided. The height of the workplace and seat should be arranged to allow alternate standing and sitting.

Design of tooling and equipment

Principles:

The hands should be relieved of all work of "holding" the workpiece where this can be done by a jig, fixture or foot-operated device.

1.

Two or more tools should be combined wherever possible.2. The hands should be relieved of work elements that can be performed by the feet.3. Multiple operations should be performed simultaneously rather than sequentially whenever possible.

4.

Where feasible, an operation should be performed on multiple parts simultaneously.5.

Equipment and controls should be designed for operator convenience and error avoidance.

6.

Manual operations should be mechanised or automated wherever feasible.7.

Having developed the improved method it is then necessary to DEFINE it so that it can always be identified as the approved way of performing the operation. This is done with a Standard Operating Procedure (SOP), which should include:

Procedure: Description of the work elements to perform the task• Tools: a list of the tools and gauges used at each stage in the procedure• Layout: plan or photograph of the workplace arrangement• Checks: checks or inspections required during the procedure• Set-up information•

Standard Operating Procedure

The aim of Standard Operating Procedures is to ensure that tasks are always performed in the same way. This will:

Reduce variations in operation time• Reduce variations in quality• Provide a basis for training new operators• Allow variations to be easily identified• Form a fixed basis for further improvement•

Define

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INSTALL it as standard practice• MAINTAIN that standard practice by regular checks•

We will not address these issues in this course but note that they are CRITICAL to the success of the process. These stages may in fact be the most difficult of all for they rely on getting people, workers and management alike to accept and implement change.

Install & maintain

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Equipment failures can lead to:

Accidents - possible harm to operators, public, other equipment• Loss of production/service - cost of lost sales• Poor quality products/services• Wasted money to replace failed equipment• Possibility of further damage during repair• Affect employee morale• Inability to meet customer demand• Affect company image•

Reliability of a system is defined as the probability that the system will perform properly for a specified period of time under a given set of operating conditions.

Clear-cut criterion for failure (and understanding of different failure modes)• Function of time (different interpretations of 'time')• Clear understanding of 'operating environment'•

What are the causes of poor reliability?

Natural wear and tear• Poor design

Poor material selection○

Incorrect understanding of operating environment○

Incomplete understanding of risks○

•

Poor manufacture and assembly• Poor maintenance• Improper operation

Untrained operators○

Operating out of rated load range○

Malice○

•

Equipment failure

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It is critical to collect data about equipment failures:

100 microprocessors are put into an oven for a dynamic stress (voltages varied) at an elevated temperature.

•

The planned test duration is 1000 hours.• Units are removed from the ovens and:

Tested every 100 hours.○

Non-failed units are returned to stress.○

•

The histogram shown is called the 'lifetime distribution' model. This shows the number of items failed in each 100 hr interval

•

Lifetime distribution model

Area = $F(T)$ = expected proportion of the entire population that fails by time T .
 The probability that k items out of a population n will fail by time T follows a binomial distribution:

Where:

Cumulative Distribution Function (CDF) of Failures, $F(t)$:

How can reliability be measured

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So reliability is:

Reliability curve (also known as survival function) of the previous example:

Mean time to failure

Mean Time to Failure (MTTF) is an important metric for measuring reliability.

When can we “expect” a randomly picked microprocessor to fail?

If time to failure of n components are available, we can generally say:

If we know the lifetime distribution function (continuous) $f(t)$ of the failure time t , MTTF is the mean value (or “expectation”) of that function:

Relationship between MTTF and Reliability:

Mean time between failures (MTBF)

This is different to MTTF, which is the mean time to the first (and only) failure for non-repairable systems. MTBF is meaningful only for repairable systems, and is the average time between subsequent failures of the same system. MTTR is the Mean Time to Repair a system (can also be the Mean Time to Replace a failed component within the system).

Failure rate

Failure rate is another important measure in reliability analysis.

The failure rate function is the probability per unit time that a failure occurs before $t + \Delta t$ given that a failure has not occurred prior to t , the beginning of the interval.

The failure rate or hazard rate $h(t)$ is defined as the limit of the failure rate function as the interval Δt approaches zero.

We can express reliability in terms of the failure rate:

200 tool tips were tested and the failures in 1000hr intervals were:

Time interval (hrs) Failures in the interval

0-1000 100

1000-2000 40

2000-3000 20

3000-4000 15

4000-5000 10

5000-6000 8

6000-7000 7

Failures Cumulative Failures

Surviving at the start

$f(t)$ $F(t)$ $R(t)$ $h(t)$

0 - 0 0 0 200 0 0 1 0

0 -1000

100 100 100 0.5 0.5 0.5 0.5

1000 -2000

40 140 60 0.2 0.7 0.3 0.4

2000 -3000

20 160 40 0.1 0.8 0.2 0.33333333

3000 -4000

15 175 25 0.075 0.875 0.125 0.375

4000 -5000

10 185 15 0.05 0.925 0.075 0.4

5000 -6000

8 193 7 0.04 0.965 0.035 0.53333333

6000 -7000

7 200 0 0.035 1 0 1

What does the failure rate look like:

$f(t)$ $h(t)$

The bath-tub curve

The complete bathtub is not universal. It is helpful to think of it as a possible combination of different failure rate curves:

The constant failure rate model

Random failures that give rise to the constant failure rate model are the most widely used basis for describing reliability. The assumption here is that the rate at which the system fails is independent of its age.

Even though a system's failure rate may change over time, we can sometimes use a constant failure rate that envelopes this curve, which may make the model slightly pessimistic.

Keep in mind the trade-off between accuracy and modelling effort •

The exponential distribution:

Because of its constant failure rate property, the exponential distribution is an excellent model for the long flat "intrinsic failure" portion of the bathtub curve. Since most components and systems spend most of their lifetimes in this portion of the Bathtub Curve, this justifies frequent use of the exponential distribution (when early failures or wear out is not a concern).

The Weibull distribution:

Reliability modelling

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The Weibull is a very flexible life distribution model with two parameters.

η (or α) scale parameter (the characteristic life)

β shape parameter

Depending on the value of the shape parameter, the Weibull model can empirically fit a wide range of data histogram shapes.

Censoring of data

Complete data: Data for which we know the exact failure time

Data with right censoring: Data for items that have not yet failed.

Data with left censoring: Data for items that failed before the start of the test.

Data with interval censoring: Data is when the exact failure time is not known but the lower and upper bounds of an interval surrounding the failure are known.

Time to failure data for a particular type of machine tool was collected for reliability analysis by observing 20 failures, and it revealed a constant failure rate of one failure approximately every 12 days of continuous operation. What is the probability that a new tool will complete an operation that requires 5 hours' time without failing?

Time-dependent failure rates

The effect of accumulated operating time T_0 on the probability that a device can survive for an additional time t .

Where:

Time elapsed at failure since the device was new

Using conditional probability, we can write:

Time to failure data for a particular type of machine tool was collected for reliability analysis by observing 20 failures, and it revealed a constant failure rate of one failure approximately every 12 days of continuous operation. What is the probability that a new tool will complete

an operation that requires 5 hours' time without failing? What if that tool had already been used for 14 days?

$$\lambda = 0.00347/\text{hr}$$

Burn-in

The study of burn-in requires us to understand time-dependent failure rates, i.e., the effect of accumulated operating time T_0 on the probability that a device can survive for an additional time t .

Don't burn-in if $\beta \geq 1$ An important assumption for effective burn-in is that component has a decreasing failure rate

($\beta < 1$).

Determining burn-in time:

Minimise probability of failure during a mission of fixed duration • Achieve a prescribed reliability for surviving units • Minimise overall expected cost (burn-in + warranty) •

Optimal burn-in time [cost model]

Cost per unit time per item for burn-in test

Cost per failure during burn-in

Cost per failure when operational $\gg C_f$

Length of burn-in testing

Operational life of the system (e.g., warranty period)

Number of units for burn-in

Cost of burn in:

Cost of failure during burn in:

Cost of failure during operation:

Expected cost including burn-in:

Expected cost per unit including burn-in:

We need to minimise $E[C]$

Maintenance

A reliability block diagram is a graphical representation of the components of a system and how they are reliability-wise related.

This may differ from how the components are physically connected.

Components Failure Survival

Series If one of the components fail If ALL the components survive

Parallel If ALL the components fail If one of the components survive

Series systems

System fails if one of the components fail. In other words, system survives if ALL the components survive.

Where:

Number of components

Reliability of the i th component

System reliability product

Critical assumption: The failures of the components are independent of each other!

Assume that the lifetime of the i th component is exponentially distributed with parameter λ_i .

System reliability is given by:

What this means is that the lifetime of the system is also exponentially distributed with parameter:

The MTBF of the system in series (assuming instant repair) is:

Reliability of systems

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Redundant systems

Let A be the event of failure of component 1 Let B be the event of failure of component 2
The system fails if both components fail.

The system failure probability is the intersection of events A and B:

Assuming that either one of the chains is strong enough to hold the load, the probability of system failure is:

System fails if ALL the components fail. In other words, system survives if one of the components survive.

Reliability of k-out of-n independent components

Three hard drives in a computer system are configured reliability-wise in parallel. At least two of them must function in order for the computer to work properly. Each hard drive is of the same size and speed, but they are made by different manufacturers and have different reliabilities. The reliability of HD #1 is 0.9, HD #2 is 0.88 and HD #3 is 0.85, all at the same mission time. Calculate the reliability of the computer system at the mission time.

Since at least two hard drives must be functioning at all times, only one failure is allowed.

This is a 2-out-of-3 configuration

-

The following combinations are possible for system operation:

All 3 hard drives operate.1. HD #1 fails while HDs #2 and #3 continue to operate.2. HD #2 fails while HDs #1 and #3 continue to operate.3. HD #3 fails while HDs #1 and #2 continue to operate.4.

-

The system reliability can now be expressed as:

-

For comparison, if only one HD was required, the system reliability would have been 99.82%

-

The number of components is n, out of which k components are necessary for effective functioning of the system.

All the components have the same reliability R • n – k redundant components • We can apply binomial distribution for evaluating the reliability of the system. •

Remember: The probability that exactly k out of the n components will survive is given by:
Hence the reliability of the system R_s (probability of survival of at least k components) is given by:

Consider a piping system having 5 pipes connected in parallel as shown in the figure.

Assume that all pipes are identical and independent. If reliability of smooth flow of the liquid from each pipeline is 0.80 for a mission time of 1 year, evaluate the reliability of the system working successfully. The system is said to work successfully if at least 3 pipelines perform their intended function successfully.

Reliability of the system is given by ($n = 5$; $k = 3$):

For reference, the reliability of the system with different k values are:

$k = 1$ 0.9997 Full redundancy

$k = 2$ 0.9933

$k = 3$ 0.9421

$k = 4$ 0.7373

$k = 5$ 0.3277 No redundancy (series)

In warm/cold standby redundancy, the components are connected in parallel but do not start operating simultaneously from the beginning of the operation of the system.

Warm standby

When the failure rate of the standby component is less in quiescent mode than in active mode

Cold standby

When the failure rate of the standby component is zero in quiescent mode (i.e. the component cannot fail when in standby)

The function of the changeover device is to sense the failure of normally operating component and in case of a failure, to bring a standby component into the normal operating mode.

What factors do we need to consider when calculating system reliability?

Need to consider the failure rate of the redundant component(s) while in passive mode and active mode separately.

-

Calculating system reliability involves estimating the conditional failure probabilities of the redundant component(s) given the failure of the active component.

-

Need to consider the reliability of the switch (i.e., the probability that the switch may not be functioning when necessary) – imperfect switching

-

May also need to consider any switching/startup time required for cold standby component to be brought online – imperfect switching

-

a. b.

The greatest reliability is achieved in the redundant configuration if we duplicate the component that is least reliable. This rule of thumb can be generalised to systems with any number of non-redundant components.

High level redundancy Low level redundancy

Workers are paid for their time on the job (even if they are not paid on an hourly rate basis).

The labour content (cost of labour time) is often a major factor in the total cost of a product or service. For any organization, it is important to know how much time will be required to accomplish a given amount of work.

Functions of time standards:

They define a “fair day’s work”• They provide a means to convert workload into staffing and equipment needs• They allow alternative methods to be compared objectively• They provide a basis for wage incentives and evaluation of worker performance• They provide time data for:

Production planning and scheduling○

Cost estimating○
Material requirements planning○

•

Factors that must be standardized before a time standard can be set:

The average worker is a worker who is representative of the persons who usually perform tasks similar to the task being measured:

If the work is performed mostly by men, then the average worker is male• If the work is performed mostly by women, then the average worker is female•

Assumed to have learned the task and is practiced and proficient at it. Well into the learning curve.

Standard performance is a pace of working that can be maintained by an average worker throughout an entire work shift without harmful effects on the worker's health or physical well-being.

Time standards

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The analyst judges the performance rating or pace of the worker relative to the definition of standard performance used by the organization:

Standard performance

Slower pace than standard

Faster pace than standard

Basic (or normal) time:

An analyst observes a worker carrying out a task. The worker does the job in 5 minutes.

According to the analyst's judgement, the worker is 'below standard' (maybe he is a newbie) and rates him at 80% of standard performance. Hence, he sets the 'basic/normal time' to do the job at:

Therefore, a standard worker should be able to do the job in 4 minutes.

Standard Method:

Must include all of the details on how the task is performed, including:

Procedure - hand and body motions○

Tools○

Equipment○

Workplace layout○

Irregular work○

Working conditions○

Setup○

•

Standard work units:

The time needed to process the work unit depends on its starting condition

Therefore this condition must be specified○

If the actual condition deviates from the specification, then extra time may be required to accomplish the task

•

•

Exactly what changes are made in the work unit by the task? • What is the final state of the completed work unit? •

Allowances

Two approaches are used by companies:

Scheduled rest breaks during the shift

Typical - one 15-minute break in mid-morning and another in mid-afternoon ○

Mainly used where worker interaction is essential ○

•

OR

An allowance is added to the basic time

This allows the worker to take breaks as required ○

Only where there is minimal interaction among workers ○

•

NOT BOTH! LUNCH TIME DOES NOT COUNT TOWARDS BREAKS!

allowances is to account for fatigue, to recognise that workers will not spend the whole shift working, and to cope with other contingencies.

Fixed Allowances:

Personal time

Toilet breaks, getting drinks, (6% typical) ○

•

Basic Fatigue

Rest allowance to overcome general fatigue (4% typical) ○

•

Variable Allowances:

These are given when the working conditions vary from those which are assumed in the basic allowances. There are three main areas which these allowances generally cover.

•

Physical Strains

Force exerted ○

Posture ○

Vibration ○

Short cycle ○

Restrictive clothing ○

•

Mental Strains

Concentration/ anxiety Monotony ○

Eye strain ○

Noise○

-

Nature of Working Conditions

Temperature and humidity Ventilation, Fumes○

Dust, Dirt○

Wet○

-

Sawing wood, outdoors, dry, autumn temp 15°, assume sawing force = 15lbf, wearing leather gloves. What should be the allowance?

Physical Force (medium): 20 Posture: 10 Vibration: 2 Short cycle: 10 Restrictive gloves: 5

Mental Concentration: 2 Monotony: 3

Working conditions

Dust: 2

Total 54

Allowance 26%

Other allowances:

Contingency Allowance

Additional allowance due to a problem with the task (e.g., raw material problem) - not greater than 5%

-

Temporary basis – solve the underlying problem○

-

Training allowance – for teaching new workers• Learning allowance – learning a new task• allowance).

The average observed time for a repetitive work cycle of packing boxes in a direct time study was 3.27min. The worker performance was rated by an analyst at 90%. The company uses an allowance factor of 13%. What is the standard time for this task?

• • •

How many boxes can the company expect a standard worker to pack in a 8 hour working day with a 30min lunch break?

Therefore, a standard worker can pack:

Think about output-based wage schemes (e.g., 45p/box). How much money will a standard worker make per hour? How much money will the observed worker make per hour?

Direct time study

Technique for measuring the times and rates of working for elements of a specified job. It is used to establish the time necessary for the job at a defined level of performance.

Direct and continuous observation of a task to record the time taken to accomplish the task

-

An appraisal of the worker's performance level is made to obtain the basic time for the task

•

The data is used to compute a standard time for the task•

Procedure: If establishing a standard, then it is essential that there has been a method study first1. Obtain and record basic information about the part/operation, the details of the process Equipment used, jigs, tools, speeds & feeds etc., the operator, the working environment.

a. 2.

Define and document the standard method3. Divide the task into work elements4. Over a number of cycles, time the work elements and calculate the mean value to obtain the observed time:

5.

Evaluate worker's pace relative to standard performance to obtain basic time (or normal time):

6.

Apply allowance factor to compute standard time:

7.

Reasons for breaking operation into elements:

The analysis of the operation by elements may show slight variations in method that could not be detected so easily from an over-all study.

•

An operator may not work at the same tempo throughout the cycle. An elemental time study permits separate performance ratings to be applied to each element of the job

•

Different elements may require different allowances•

Guidelines for defining work elements:

Each work element should consist of a logical group of motion elements• Beginning point of one element should be the end point of the preceding element• Each element should have a readily identifiable point• Work elements should never be too long or too short• Irregular work elements should be identified and distinguished from regular work elements

•

Manual elements should be separated from machine elements• Internal elements should be separated from external elements•

Timing techniques

Snapback timing Continuous timing

Watch is started at the beginning of every work element by snapping it back to zero at the end of the previous element.

•

The analyst must note and record the final time for that element just as the watch is

•

Watch is zeroed at the beginning of the first cycle and allowed to run continuously throughout the duration of the study.

-

The analyst records the running time on the stopwatch at the end of each respective

-

Measuring time taken

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being zeroed

Advantages:

The analyst can readily see how the element times vary from one cycle to the next

-

No subtraction is necessary to obtain individual element times

-

- element.

Advantages:

Elements are not omitted by mistake • Manipulation of the stopwatch is minimised

-

-

A direct time study was performed for a task. The regular cycle consisted of three elements, a, b, and c. Element d is an irregular element performed every five cycles.

Work element a b c d

Observed time (min) 0.56 0.25 0.50 1.10

Performance rating 100% 80% 110% 100%

Determine basic time and standard time for the cycle using an allowance factor of 15%. Basic time:

Standard time:

Measurement problems:

How can we be sure that the limited “sample” of times observed is actually representative of the truth?

Is the mean of the observed times equal to the true value of the time to perform the task?

-

Unless you make a very large number of observations, you cannot make that claim.

-

-

However, the number of observations that can be made during a direct time study is

-

limited (cost, time, effort)

So, the question is how many observations must be made so that we are 'fairly' confident about the outcome of the study?

-

Confidence

Precision is concerned with the expected variability within a single time study. Precision of a time standard is determined at a certain confidence level.

For example, the standard time calculated based on a time study for a task is 4.00 min, and we are 98% confident that the actual time is within 5% of that time

-

To cope with variations, we need to time a number of repetitions (n) of the job, from which we can calculate a confidence interval for the mean value (μ) of the time. We assume that the observed work element times are normally distributed about the true value of the time. After several cycles, we can calculate the mean from the observed data.

Probability Density Function of measured time (x):

The term:

Is set to our desired interval size (e.g., $\pm 10\%$ of)

Now, we do not know the actual standard deviation σ ! After several cycles, calculate the mean () and sample standard deviation (s) from the observed data:

Because the sample size is usually small (< 30), the statistic:

will be t-distributed with $n - 1$ degrees of freedom.

E.g., for 95% confidence, $\alpha = 0.05$

A total of 9 cycles of an assembly operation have been observed during a direct time study.

The mean time = 1.30 min, and the corresponding sample standard deviation $s = 0.20$ min.

Calculate 95% confidence limits for the assembly time.

"I am 95% confident that the actual mean time of the assembly operation will be somewhere between 1.146 to 1.454 minutes" "I am 95% confident that the assembly operation will take somewhere between 1.146 to 1.454 minutes"

By rearranging the formula, we can find how many observations we need to take to get within certain error limits at a given level of confidence. If we define k (e.g., $k = 10\%$ or 0.1) as the max fractional error in the average measured time that we are prepared to accept.

Then the number of observations to take is:

Remember the practicality/cost vs. accuracy trade-off!

A time study analyst has collected 10 readings on a particular work element of interest and would like to consider how many more cycles to time. Based on the sample, the mean time for the element is 0.40 min and the sample standard deviation is 0.07 min. At a 95% confidence level, how many cycles should be timed to ensure the actual element time is within $\pm 10\%$ of the mean?

6 more cycles

A reduction in cycle time that occurs in a repetitive work activity as the number of cycles increases. When a worker accomplishes a task over and over, the time required for each successive cycle decreases as he or she learns the task:

At first the learning effect is rapid, and the cycle time decreases significantly• As more and more cycles are completed, the cycle time reduction becomes less and less•

A property of this curve is that every time X doubles, Y decreases to a fixed %, say p% of its previous value. The curves are often referred to as "p % learning curves".

Learning curve theory

Learning rate = the proportion by which the task time is multiplied every time the number of task cycles doubles Example: If $T_1 = 10$ hr and $LR = 80\%$, then:

• • • and so on•

When learning curve time data are plotted on log-log coordinates, the plot yields a straight line with slope m. Relationship between m and LR:

Learning curves

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Higher LR means slower learning Same Learning Curve in Log-Log Plot:

Type of work LR %

Assembly 85

Prototype assembly 65

Clerical ops 75-85

Inspection 85

Machining 90

Welding 85-90

Factors affecting learning curves

Self Organisation

Practice - Worker becomes familiar with the task - (moves from conscious → sub-conscious)

•

Worker makes fewer mistakes as the task is repeated• Hand and body motions become more efficient, and there is a rhythm and pattern developed – strengthening of muscles

•

Minor adjustments in workplace layout to reduce distances – better tools and equipment

•

Methods improvements• Fine-tuning of machinery and tooling

•

Development of special tooling

•

Technological improvements

•

Product design improvements

-

Leadership and better motivation of workers

-

Factors impeding learning:

Physical and mental limits• Frustration• Inertia• Interruption / forgetting• Constraints imposed by other members of group• Shortage of resources, lack of investment in new technology• Lack of Leadership• Low motivation of workers•

Learning curve effect tends to be disrupted by interruptions in production. For example due to:

Batch production, in which there are extended periods between repeat orders• Vacations•

Raw material shortages•

Product complexity

More opportunities for learning - higher T1 and lower LR○

-

Preproduction planning

Fewer opportunities for learning - lower T1 and higher LR○

-

Labour turnover - slower learning (higher LR)• Plateau model - learning curve consists of two phases:

Start-up - log-linear model applies1. Steady state - no further improvement2.

-

Why the Plateau Model occurs:

In a worker-machine system, worker is finally limited by the speed of the machine• No incentives for workers to make further improvements• Management is unwilling or unable to invest in newer production technologies• Management does not believe that learning can continue indefinitely•

Therbligs

17 basic motion elements

Catalogued by Frank Gilbreth in the early part of the Twentieth century• Basic building blocks of virtually all manual work performed at a single location• With modification, used today in several work measurement systems•

Effective Therbligs Ineffective Therbligs

Physical basic motion elements

Transport empty (TE) Grasp (G) Transport loaded (TL) Release load (RL) Use (U) Assemble (A) Disassemble (DA)

Hold (H) Preposition (PP)

Physical and mental basic motion elements

Position (P) Search (Sh) Select (St)

Mental basic elements Inspect (I) Plan (Pn)

Delay elements Rest (R) Unavoidable delay (UD) Avoidable delay (AD)

Micromotion analysis

Analysis of therbligs that make up a repetitive task. Objectives:

Eliminate ineffective therbligs if possible• Avoid holding objects with hand

Use workholder○

•

Combine therbligs

Perform right-hand and left-hand motions simultaneously○

•

Simplify overall method• Reduce time for a motion, e.g., shorten distance•

Predetermined motion time systems

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Predetermined Motion Time Systems

PMTS is a database of basic motion elements and their associated basic/normal time values, together with procedures for applying the data to analyse manual tasks and establish standard times for the tasks. Basic motions include:

Reach• Grasp• Move• Release •

Procedure: Synthesize method that would be used to perform the task – The method is described in terms of basic motion elements

1.

Retrieve basic time values for each motion element – Sum the element times to determine the task basic time

2.

Evaluate method to make improvements3. Apply allowances to determine standard time4.

Methods-Time Measurement

“Procedure which analyses any manual operation or method into the basic motions required to perform it and assigns to each motion a predetermined time standard which is determined by the nature of the motion and the conditions under which it is made”

Time units are TMUs

MTM is a family of products available through the MTM Association in Des Plaines, Illinois.

MTM-1 Motion Elements:

Reach• Grasp• Move• Position• Release• Disengage• Turn• Apply pressure• Eye travel, eye focus, and reading•

Body, leg and foot motions• Simultaneous motions•

Other MTM Systems:

MTM-2 – Second-level PMTS in which basic motion elements are combined into motion aggregates

GET – combines Reach and Grasp○

PUT – combines Move and Position○

•

MTM-3 – Third-level PMTS which has four motion categories

Handle1. Transport2. Step and foot motions3. Bend and arise4.

-

Advantages of predetermined time standards:

The detailed systems produce a more detailed description of the work than other methods.

-

As each basic motion is described along with its variables, so alternative methods can be readily compared using time as a criterion.

-

Methods can be developed and standards established before production operations start. • PTS do not require the assessment of rate of working and the use of the stop watch as is required in time study. This can remove one of the impediments to good industrial relations.

-

PTS are objective and consistent. •

A SDS is a database of basic/normal time values, usually organized by work elements, that can be used to establish time standards for tasks composed of work elements similar to those in the database.

Basic times in the database are usually compiled from previous direct time studies. • Using a standard data system, time standards can be established before the job is running.

-

A SDS Database is a catalogue of basic times, organized to allow the analyst to access the values corresponding to work elements performed under various work variables:

Work variables - factors that affect the basic time of the element

Work unit characteristics (e.g., size, weight)○

Task parameters (e.g., distances moved)○

Working conditions (e.g., illumination)○

-

Various formats can be used in the SDS database:

Charts • Tables • Mathematical formulas • Worksheets • Computerized database and retrieval system •

When a SDS is most suitable:

Similarity in tasks

With many similar tasks, it is more efficient to use an SDS than direct time study○

-

Large number of standards to be set

More productive than direct time study○

-

Need to set standards before production

Direct time study requires observation of actual task to set standard, SDS does not○

-

Advantages Disadvantages

Standard data systems

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standards

Capability to set standards before production

-

Avoids need for performance rating• Consistency in the standards •

Developing a SDS requires considerable time and cost

-

Source of data

Large file of previous detailed time study data must exist

-

-

Risk of improper applications

Attempting to set standard for tasks not covered by SDS

-

-

Work Sampling is a statistical technique for determining the proportions of time spent by subjects in various defined categories of activity.

Subjects = workers, machines• Categories of activity = setting up a machine, producing parts, idle, etc.• For statistical accuracy:

Observations must be taken at random times○

Period of the study must be representative of the types of activities performed by the subjects

-

-

When is work sampling appropriate?

Sufficient time is available to perform the study

Several weeks usually required for a work sampling study○

-

Multiple subjects

Work sampling suited to studies involving more than one subject○

•

Long cycle times for the jobs covered by the study• Nonrepetitive work cycles
Jobs consist of various tasks rather than a single repetitive task○

•

Work sampling applications:

Machine utilization - how much time is spent by machines in various categories of activity

•

Worker utilization - how workers spend their time• Allowances for time standards -
assessment of delay components in allowance• Average unit time - determining the average
time on each work unit• Time standards - limited statistical accuracy when standards set by
work sampling•

Steps in a work sampling study: Define the problem.1. Determine the required accuracy and
required confidence limits.2. Make a preliminary estimate of the % occurrence of the desired
activity. Usually done by carrying out a pilot study.

3.

Design the main study. Find number of observations required, design sampling plan with
randomised times, determine number of observers.

4.

Carry out study according to above plan, summarising the data at the end of each shift.5.

A total of 500 observations taken at random times during a one-week period (40 hours) on 10
machines with results shown below.

Activity (work) sampling

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Category No. of observations

(1) Being set up 75

(2) Running production 300

(3) Machine idle 125

Total 500

How many hours per week did an average machine spend in each category? Proportions of
time determined as number of observations in each category divided by 500. Time in each
category determined by multiplying proportion by total hours (40 hr).

Category Proportion Hrs per category

(1) Being set up

(2) Running production

(3) Machine idle

Total

But how accurate is this?

If we had carried out another study would we have got a similar result?• Activity sampling is
essentially drawing a sample of size n from an infinite population, and from the sample
making inferences about the parameters of the underlying population.

•

We can use statistics to quantify sampling errors.•

Statistical basis of work sampling

Binomial distribution, in which parameter p = true proportion of time spent in a given category of activity. There are usually multiple activity categories, so we have: proportions for K different activity categories The binomial distribution can be approximated by the normal distribution (providing the number of observations n is greater than 30)

In a sampling study:

the proportion of the total number of observations devoted to an activity category of interest

The proportion is our estimate of the true value of the population proportion p [Need to ensure there is no bias!]. The general statement of a confidence interval for relative to p can be expressed as follows:

Where:

This can be rearranged as follows:

The true value of p lies between:

And:

At a confidence level of:

For 95% confidence, $\alpha = 0.05$

Confidence interval Calculate the 95% confidence interval for the proportion of running time for the machines in the previous example.

Category Proportion Hrs per category

(1) Being set up

(2) Running production

(3) Machine idle

Total

Proportion of running time:

Standard deviation:

Confidence interval = 95%

How many observations are required to achieve a given confidence interval about the estimate of p ? We need to decide two parameters:

Confidence level $1 - \alpha$ This allows us to find the corresponding value of $z_{\alpha/2}$.

1.

The half-width c of the confidence interval, defined as the desired acceptable deviation

2.

from p Thus, we have $p \pm ca$.

Given $z_{\alpha/2}$ and c , the number of observations required to achieve the specified confidence level is given by the following:

Based on our defined values of $z_{\alpha/2}$ and c , we can determine the associated value of that achieves the desired confidence interval:

In terms of n :

Therefore, given $z_{\alpha/2}$ and c , the number of observations required to achieve the specified confidence level is given by the following:

Determine how many observations will be required to estimate the proportion of time used to set up the 10 machines. The confidence interval must be within ± 0.03 of the true proportion, which the line manager initially estimates to be . A 95% confidence level will be used.

At the 95% confidence level:

The confidence interval is:

So the half-width of the interval:

Advantages Disadvantages

Can be used to measure activities that are impractical to measure by direct observation

-

Multiple subjects can be included• Requires less time and lower cost than continuous direct observation

-

Training requirements less than DTS or PMTS• Less tiresome and tedious on observer than continuous observation

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Being a subject in work sampling is less demanding than being watched continuously for a long time

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Not as accurate for setting time standards as other work measurement techniques

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Usually not practical to study a single subject

-

Work sampling provides less detailed information about work elements than DTS or PMTS

-

Since work sampling deals with multiple subjects, individual differences will be missed

-

Workers may be suspicious because they do not understand the statistical basis of work sampling

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Behaviour of subjects may be influenced by the act of observing them